

- 2(a)** By Newton's 2nd law, M1

$$T - mg \sin \theta = ma$$

 At equilibrium, M1

$$T_1 = ke = mg \sin \theta$$

$$T - T_1 = ma$$
 A1

$$k(x+e) - ke = ma$$

 Since x is measured downwards from equilibrium,

$$a = -\frac{kx}{m}$$
- (b)** Acceleration is directly proportional to displacement from equilibrium and the B1
 acceleration is in the opposition direction to displacement (or directed towards B1
 equilibrium).
- (c)**
$$v = \pm \omega \sqrt{x_o^2 - x^2}$$

$$v_o = \omega x_o$$

$$\frac{v}{v_o} = \frac{\sqrt{x_o^2 - x^2}}{x_o}$$
 M1

$$0.25 = \frac{\sqrt{3^2 - x^2}}{3}$$
 A1

$$x = 2.9 \text{ cm}$$
- (d)(i)** Resonance is illustrated. The driving frequency of the oscillator is equal to the natural B1
frequency of the mass-spring system, maximum energy is transferred to the system B1
 and the mass-spring system oscillates with maximum amplitude.
- (d)(ii)** $|a| = \omega^2 x_o = (2\pi f)^2 x_o = [2\pi(1.2)]^2 (0.06) = 3.4 \text{ ms}^{-2}$ A1

Comments

Most students obtained the correct answers for (d)(i)(ii).